

1. we establish a connection using mavlink with the rovr. No issue here.
  2. We create a fake goal using the Fake joystick class. No issue here.
  3. We start the imu\_listener that will collect data from the rovr. No issue here.
  4. We create the Rov\_gym\_environment. No issue here.
  5. During environment resets, we set "self.state\_history = [state] \* self.history\_length". This should not be an issue, it is as if the rovr was stationnary for a while.
  6. We create the agent using DeterministicGCAgent. This then creates an actor and a Critic, both based on a GRU Network. They hae the same sizes appart from inputs and outputs.
  7. If needed, we switch the goal (during exploration and first learning phase)  
then, inside the step loop : {
  8. We select an action based on the phase (random or actor choice)  
8.1) select\_action takes a state, and outputs the action, after feeding the state to the actor.
  9. Environmment step, where we apply an action for 0.1s, then compute the next state\_sequence and the reward for our action.
  10. We convert observations and action to an appropriate format before pushing them in the replay\_buffer.
  11. Once in every "update\_every", we update either the critic only (exploration-specific) or the critic and the actor.  
}
- and we repeat 8-11 until convergence while saving occasionally.